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Course: Biology 1002

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R Assignment I.

Q1.
```
setwd("~/Documents/R/BIO_R_ASS_2")
> install.packages("dplyr")
```

```
WARNING: Rtools is required to build R packages but is not currently instal
led. Please download and install the appropriate version of Rtools before p
roceeding:


https://cran.rstudio.com/bin/windows/Rtools/
Installing package into ‘C:/Users/gephe/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
also installing the dependencies ‘utf8’, ‘pkgconfig’, ‘withr’, ‘cli’, ‘gen
erics’, ‘glue’, ‘lifecycle’, ‘magrittr’, ‘pillar’, ‘R6’, ‘rlang’, ‘tibble’
, ‘tidyselect’, ‘vctrs’trying URL 'https://cran.rstudio.com/bin/windows/co
ntrib/4.5/utf8_1.2.6.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/pkgconfig_2.0
.3.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/withr_3.0.2.z
ip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/cli_3.6.5.zip
'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/generics_0.1.
4.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/glue_1.8.0.zi
p'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/lifecycle_1.0
.5.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/magrittr_2.0.
4.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/pillar_1.11.1
.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/R6_2.6.1.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/rlang_1.1.7.z
ip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/tibble_3.3.1.
zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/tidyselect_1.
2.1.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/vctrs_0.7.1.z
ip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/dplyr_1.2.0.z
ip'
package ‘utf8’ successfully unpacked and MD5 sums checked
package ‘pkgconfig’ successfully unpacked and MD5 sums checked
package ‘withr’ successfully unpacked and MD5 sums checked
```

```
package ‘cli’ successfully unpacked and MD5 sums checked
package ‘generics’ successfully unpacked and MD5 sums checked
package ‘glue’ successfully unpacked and MD5 sums checked
package ‘lifecycle’ successfully unpacked and MD5 sums checked
package ‘magrittr’ successfully unpacked and MD5 sums checked
package ‘pillar’ successfully unpacked and MD5 sums checked
package ‘R6’ successfully unpacked and MD5 sums checked
package ‘rlang’ successfully unpacked and MD5 sums checked
package ‘tibble’ successfully unpacked and MD5 sums checked
package ‘tidyselect’ successfully unpacked and MD5 sums checked
package ‘vctrs’ successfully unpacked and MD5 sums checked
package ‘dplyr’ successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\gephe\AppData\Local\Temp\RtmpKMHUHU\downloaded_packages
> library(dplyr)

Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:


    filter, lag


The following objects are masked from ‘package:base’:


    intersect, setdiff, setequal, union


Warning message:
package ‘dplyr’ was built under R version 4.5.2

#Q2. moosedata <- read.csv("MoosePopulation.csv")
#Q3
> View(moosedata)
>
> moose_clean <- na.omit(moosedata)
>
> View(moose_clean)
> #Q4
> moose_sel <- select(moose_clean, Ecoregion, Year, Area, Estimated_Moose_
Pop)
> #Q5
> year_min <- min(moose_sel$Year)
>
> year_min
[1] 1904
>
> moose_max <- max(moose_sel$Estimated_Moose_Pop)
>
> moose_max
[1] 41250
> #Q6
> moosedata2 <- mutate(moose_sel,
+                      MooseDensity = Estimated_Moose_Pop / Area)
```

```
>
> moosedata2 <- mutate(moose_sel, MooseDensity = Estimated_Moose_Pop / Are
a)
> View(moosedata2)
> #Q7
> plot(moosedata2$Year,
+      moosedata2$MooseDensity,
+      type = "l",
+      xlab = "Year",
+      ylab = "Moose density (moose per km^2)",
+      main = "Changes in moose density over time")
> #Q8
> moose_west <- filter(moosedata2, Ecoregion == "Western_Forests")
> View(moose_west)
> #Q9
> moose_2020 <- filter(moosedata2, Year == 2020)
>
> View(moose_2020)
>
> moose_2020_high <- filter(moose_2020, MooseDensity > 2.0)
>
> View(moose_2020_high)
>
> moose_2020_high_byD <- arrange(moose_2020_high, desc(MooseDensity))
>
> View(moose_2020_high_byD)
> #Q10
> moosefinal <- moosedata2 %>%
+     filter(Year == 2020) %>%
+     filter(MooseDensity > 2.0) %>%
+     arrange(desc(MooseDensity)) %>%
+     print()
                   Ecoregion Year  Area Estimated_Moose_Pop MooseDensity
1 Northern_Peninsula_Forests 2020 12000               32000     2.666667
2        North_Shore_Forests 2020  8400               21740     2.588095
3            Western_Forests 2020  6800               17000     2.500000
4            Central_Forests 2020 20500               41250     2.012195
> #Q11
> saplings <- read.csv("SaplingStudy.csv")
>
> View(saplings)
>
> sap_clean <- na.omit(saplings)
>
> View(sap_clean)
> #Q12
> sap_reg_browse <- sap_clean %>%
+     group_by(Ecoregion) %>%
+     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
+     print()
# A tibble: 9 × 2
  Ecoregion                  AverageBrowsing
  <chr>                                <dbl>
1 Avalon_Forests                        2
2 Central_Forests                       4
3 EasternHyperOceanicBarrens            2.4
4 Long_Range_Barrens                    2.6
5 Maritime_Barrens                      1.83
```

```
6 North_Shore_Forests                    4.38
7 Northern_Peninsula_Forests             4.57
8 StraitOfBelleIsleBarrens               1
9 Western_Forests                        4.5
>
> avg_browse_reg <- sap_reg_browse %>%
+     arrange(desc(AverageBrowsing)) %>%
+     print()
# A tibble: 9 × 2
  Ecoregion                   AverageBrowsing
  <chr>                                 <dbl>
1 Northern_Peninsula_Forests             4.57
2 Western_Forests                        4.5
3 North_Shore_Forests                    4.38
4 Central_Forests                        4
5 Long_Range_Barrens                     2.6
6 EasternHyperOceanicBarrens             2.4
7 Avalon_Forests                         2
8 Maritime_Barrens                       1.83
9 StraitOfBelleIsleBarrens               1
>
> # Highest average browsing = first row
> # Lowest average browsing = last row
>
> avg_browse_reg <- sap_reg_browse %>%
+     arrange(desc(AverageBrowsing)) %>%
+     print()
# A tibble: 9 × 2
  Ecoregion                   AverageBrowsing
  <chr>                                 <dbl>
1 Northern_Peninsula_Forests             4.57
2 Western_Forests                        4.5
3 North_Shore_Forests                    4.38
4 Central_Forests                        4
5 Long_Range_Barrens                     2.6
6 EasternHyperOceanicBarrens             2.4
7 Avalon_Forests                         2
8 Maritime_Barrens                       1.83
9 StraitOfBelleIsleBarrens               1
>
> # Highest average browsing: <name from first row>
> # Lowest average browsing: <name from last row>
>
> #Q13
> sap_reg_height <- sap_clean %>%
+     group_by(Ecoregion) %>%
+     summarize(AvgHeight = mean(Height)) %>%
+     print()
# A tibble: 9 × 2
  Ecoregion                   AvgHeight
  <chr>                           <dbl>
1 Avalon_Forests                   32.4
2 Central_Forests                  23.8
3 EasternHyperOceanicBarrens       31.6
4 Long_Range_Barrens               29.9
5 Maritime_Barrens                 26.7
6 North_Shore_Forests              22.3
7 Northern_Peninsula_Forests       19.9
```

```
8 StraitOfBelleIsleBarrens        25.4
9 Western_Forests                 18.9
>
> sap_reg_height_low <- sap_reg_height %>%
+     filter(AvgHeight < 20) %>%
+     print()
# A tibble: 2 × 2
  Ecoregion                  AvgHeight
  <chr>                          <dbl>
1 Northern_Peninsula_Forests      19.9
2 Western_Forests                 18.9
>
> # Ecoregions with AvgHeight < 20 cm are considered severely browsed
>
> #Q14
> sap_spe_browse <- sap_clean %>%
+     group_by(Species) %>%
+     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
+     print()
# A tibble: 6 × 2
  Species        AverageBrowsing
  <chr>                    <dbl>
1 "Alder "                  4.25
2 "Balsam_Fir"              3.14
3 "Black_Ash"               5
4 "Black_Spruce"            2.33
5 "White_Birch"             3.14
6 "Willow"                  4.31
>
> avg_browse_spe <- sap_spe_browse %>%
+     arrange(desc(AverageBrowsing)) %>%
+     print()
# A tibble: 6 × 2
  Species        AverageBrowsing
  <chr>                    <dbl>
1 "Black_Ash"               5
2 "Willow"                  4.31
3 "Alder "                  4.25
4 "White_Birch"             3.14
5 "Balsam_Fir"              3.14
6 "Black_Spruce"            2.33
>
> # Highest browsing species = first row
> # Lowest browsing species = last row
> #Q15
> fir_reg_browse <- sap_clean %>%
+     filter(Species == "Balsam_Fir") %>%
+     group_by(Ecoregion) %>%
+     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
+     print()
# A tibble: 7 × 2
  Ecoregion                  AverageBrowsing
  <chr>                                <dbl>
1 Avalon_Forests                        2
2 Central_Forests                       3.5
3 EasternHyperOceanicBarrens            2
4 Long_Range_Barrens                    1
5 Maritime_Barrens                      2
```

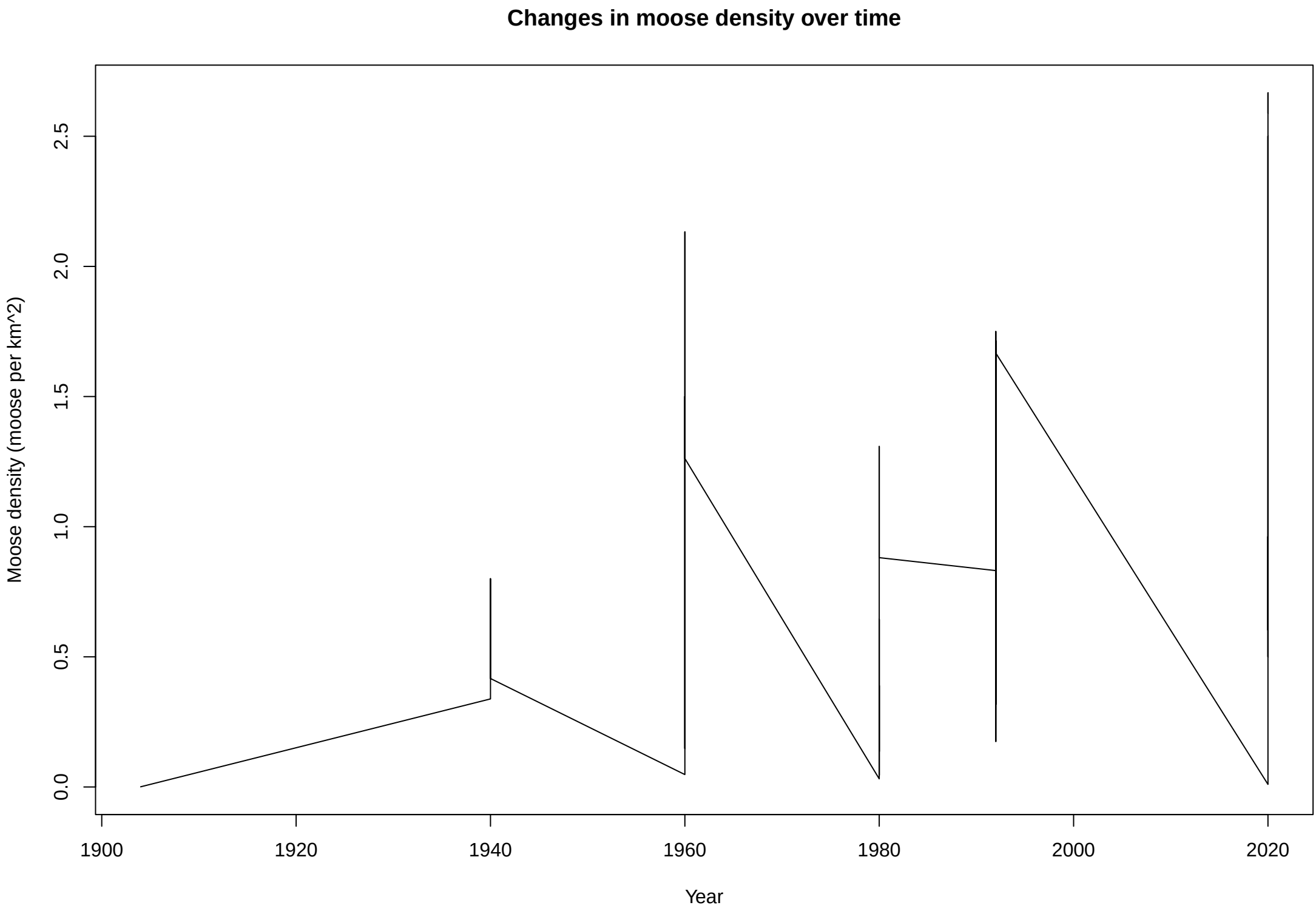
```
6 North_Shore_Forests                    4.5
7 Northern_Peninsula_Forests             4.25
> #Q16
> barplot(fir_reg_browse$AverageBrowsing,
+         names.arg = fir_reg_browse$Ecoregion,
+         xlab = "Ecoregion",
+         ylab = "Average browsing score",
+         main = "Average Balsam Fir browsing intensity by ecoregion",
+         col = "forestgreen",
+         cex.names = 0.6)
> #Q17
> #  Mean Black Spruce browsing by ecoregion
> spruce_reg_browse <- sap_clean %>%
+     filter(Species == "Black_Spruce") %>%
+     group_by(Ecoregion) %>%
+     summarize(AverageBrowsing = mean(BrowsingScore)) %>%
+     print()
# A tibble: 8 × 2
  Ecoregion                  AverageBrowsing
  <chr>                                <dbl>
1 Avalon_Forests                         0.5
2 Central_Forests                        3
3 EasternHyperOceanicBarrens             0
4 Long_Range_Barrens                     2
5 Maritime_Barrens                       0
6 North_Shore_Forests                    4
7 Northern_Peninsula_Forests             4
8 Western_Forests                        3.5
> # Bar plot for Black Spruce browsing
> barplot(spruce_reg_browse$AverageBrowsing,
+         names.arg = spruce_reg_browse$Ecoregion,
+         xlab = "Ecoregion",
+         ylab = "Average browsing score",
+         main = "Average Black Spruce browsing intensity by ecoregion",
+         col = "darkolivegreen",
+         cex.names = 0.6)
> # Comparison comment (based on my results)
> # Black Spruce generally shows (higher/lower/similar) browsing intensity
than Balsam Fir across most ecoregions, suggesting (greater/lesser/similar
) moose preference for Black Spruce.
>#Q18
> sap_reg_tally <- sap_clean %>%
+     group_by(Ecoregion) %>%
+     tally() %>%
+     print()
# A tibble: 9 × 2
  Ecoregion                      n
  <chr>                      <int>
1 Avalon_Forests                 5
2 Central_Forests                5
3 EasternHyperOceanicBarrens     5
4 Long_Range_Barrens             5
5 Maritime_Barrens               3
6 North_Shore_Forests            8
7 Northern_Peninsula_Forests     7
8 StraitOfBelleIsleBarrens       1
9 Western_Forests                5
> #Q19
```

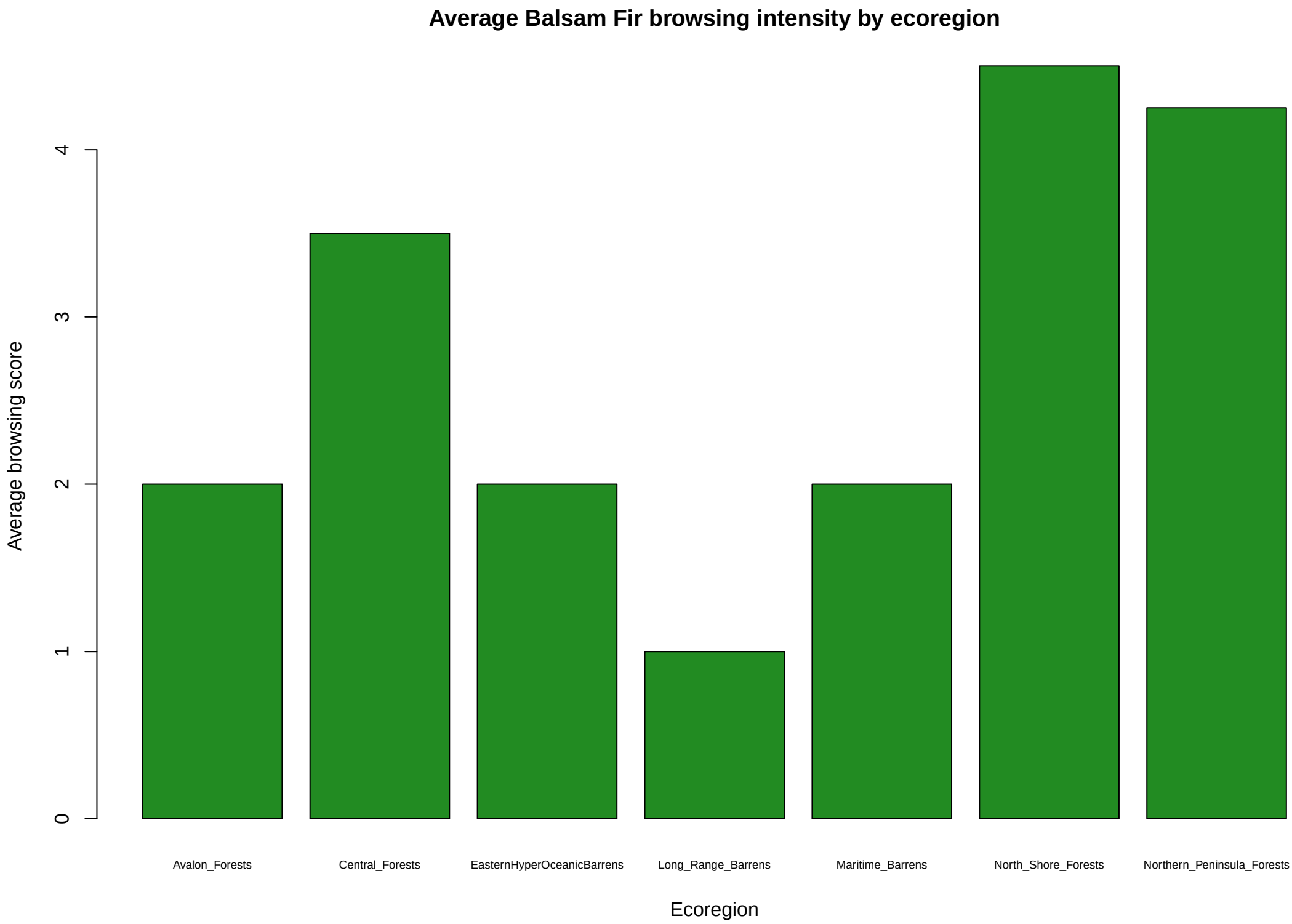
```
> sap_spe_tally <- sap_clean %>%
+     group_by(Species) %>%
+     tally() %>%
+     print()
# A tibble: 6 × 2
  Species            n
  <chr>          <int>
1 "Alder "           8
2 "Balsam_Fir"      11
3 "Black_Ash"        1
4 "Black_Spruce"     9
5 "White_Birch"      7
6 "Willow"           8
>
>
> # 20a. The SaplingStudy dataset is not evenly distributed.
> # North_Shore_Forests and Northern_Peninsula_Forests are overrepresented
,
> # while StraitOfBelleIsleBarrens and Maritime_Barrens are underrepresent
ed.
> # In terms of species, Balsam_Fir and Black_Spruce are sampled more freq
uently,
> # whereas Black_Ash is highly underrepresented with only one observation
.
>
> # 20b. Recognizing bias in ecological datasets is important because unev
en sampling
> # can distort estimates of browsing intensity and lead to misleading con
clusions
> # about moose preferences or differences among ecoregions and species.
> #Q21a
> moose_2020b <- moose_clean %>%
+     filter(Year == 2020) %>%
+     mutate(MooseDensity = Estimated_Moose_Pop / Area)
>
> View(moose_2020b)
> #Q21b
> moose_sap <- left_join(moose_2020b, sap_clean, by = "Ecoregion")
>
> View(moose_sap)
>#Q22
> sum_spe_browse <- moose_sap %>%
+     group_by(Species, Ecoregion) %>%
+     summarize(
+         AvgBrowsing = mean(BrowsingScore),
+         AvgDensity  = mean(MooseDensity)
+     ) %>%
+     print()
`summarise()` has regrouped the output.
i Summaries were computed grouped by Species and Ecoregion.
i Output is grouped by Species.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(Species, Ecoregion))` for per-operation grouping i
nstead.
# A tibble: 38 × 4
# Groups:   Species [6]
   Species      Ecoregion                    AvgBrowsing AvgDensity
   <chr>        <chr>                              <dbl>      <dbl>
```

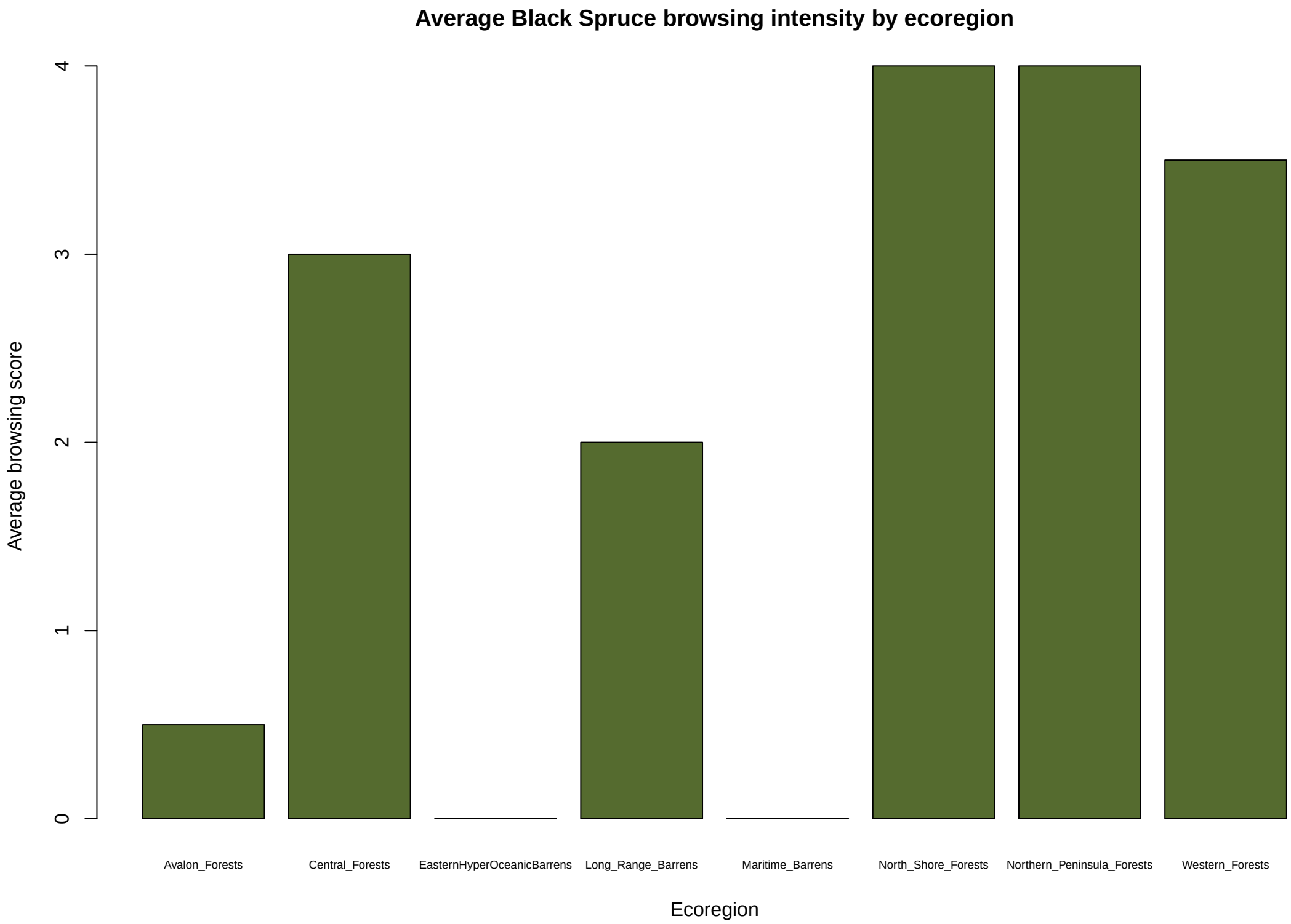
```
 1 "Alder "     Avalon_Forests                    4        0.962
 2 "Alder "     Central_Forests                   4.5      2.01
 3 "Alder "     EasternHyperOceanicBarrens        4        0.5
 4 "Alder "     Long_Range_Barrens                3.5      0.602
 5 "Alder "     Maritime_Barrens                  3.5      0.960
 6 "Alder "     North_Shore_Forests               4.5      2.59
 7 "Alder "     Northern_Peninsula_Forests        5        2.67
 8 "Alder "     Western_Forests                   5        2.5
 9 "Balsam_Fir" Avalon_Forests                    2        0.962
10 "Balsam_Fir" Central_Forests                   3.5      2.01
# ℹ 28 more rows
# ℹ Use `print(n = ...)` to see more rows
>
> View(sum_spe_browse)
>
> write.csv(sum_spe_browse, "sum_spe_browse.csv", row.names = FALSE)
> #Q23
> install.packages("ggplot2")
WARNING: Rtools is required to build R packages but is not currently instal
led. Please download and install the appropriate version of Rtools before p
roceeding:


https://cran.rstudio.com/bin/windows/Rtools/
Installing package into ‘C:/Users/gephe/AppData/Local/R/win-library/4.5’
(as ‘lib’ is unspecified)
also installing the dependencies ‘cpp11’, ‘farver’, ‘labeling’, ‘RColorBre
wer’, ‘viridisLite’, ‘gtable’, ‘isoband’, ‘S7’, ‘scales’trying URL 'https:
//cran.rstudio.com/bin/windows/contrib/4.5/cpp11_0.5.3.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/farver_2.1.2.
zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/labeling_0.4.
3.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/RColorBrewer_
1.1-3.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/viridisLite_0
.4.3.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/gtable_0.3.6.
zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/isoband_0.3.0
.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/S7_0.2.1.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/scales_1.4.0.
zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.5/ggplot2_4.0.2
.zip'
package ‘cpp11’ successfully unpacked and MD5 sums checked
package ‘farver’ successfully unpacked and MD5 sums checked
package ‘labeling’ successfully unpacked and MD5 sums checked
package ‘RColorBrewer’ successfully unpacked and MD5 sums checked
package ‘viridisLite’ successfully unpacked and MD5 sums checked
package ‘gtable’ successfully unpacked and MD5 sums checked
package ‘isoband’ successfully unpacked and MD5 sums checked
package ‘S7’ successfully unpacked and MD5 sums checked
package ‘scales’ successfully unpacked and MD5 sums checked
package ‘ggplot2’ successfully unpacked and MD5 sums checked
```

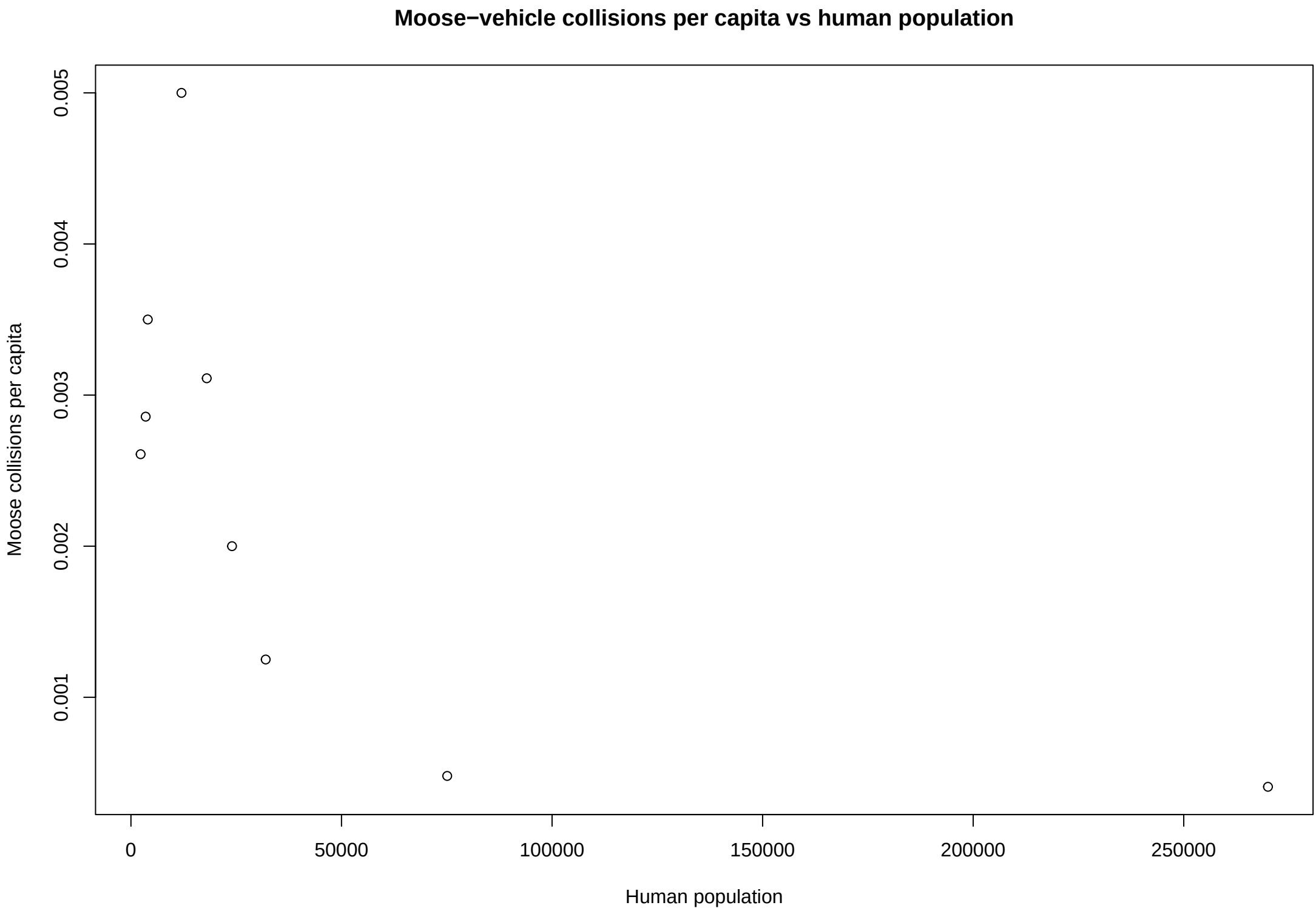
```
The downloaded binary packages are in
        C:\Users\gephe\AppData\Local\Temp\RtmpKMHUHU\downloaded_packages
>
> library(ggplot2)
Warning message:
package ‘ggplot2’ was built under R version 4.5.2 >
> ggplot(sum_spe_browse, aes(x = AvgDensity, y = AvgBrowsing, color= Speci
es)) +
+     geom_point(size = 3) +
+
+     theme_minimal() +
+     labs(title = "Browsing Intensity Across Moose Density by Species",
+          x = "Average Moose Density",
+          y = "Average Browsing Score")
> # The figure provides partial support for the researchers’ hypothesis. A
t lower
> # moose densities, browsing intensity differs noticeably among species,
whereas at
> # higher moose densities the browsing scores converge, suggesting a shif
t toward
> # more generalist feeding as density increases.
>
> # Moose appear to favour Balsam Fir and Willow, which show consistently
higher
> # average browsing scores across moose densities, while White Birch and
Black
> # Spruce tend to be browsed the least.
>
> # No sapling species is missing from the figure; however, Black Ash appe
ars only
> # once due to very limited sampling, which restricts interpretation of b
rowsing
> # patterns for that species.
> #Q24
> collisions2020 <- c(56, 60, 14, 36, 48, 10, 40, 110, 6)
> human_pop <- c(18000, 12000, 4000, 75100, 24000,3500, 32000, 270000,
2300)
> study_sites <- c("North_Shore_Forests","Northern_Peninsula_Forests", "Lo
ng_Range_Barrens","Central_Forests","Western_Forests","EasternHyperOceanic
Barrens","Maritime_Barrens","Avalon_Forests","StraitOfBelleIsleBarrens")
> #Q25
> moose_coll <- data.frame(collisions2020, human_pop, study_sites)
> View(moose_coll)
>
> # Rename study_sites column to Ecoregion
> moose_coll2 <- moose_coll %>%
+     rename(Ecoregion = study_sites)
>
> # Join with moose_2020 dataset
> coll_merge <- left_join(moose_2020, moose_coll2, by = "Ecoregion")
>
> View(coll_merge)
>#Q26
> plot(coll_merge$MooseDensity,
+      coll_merge$collisions2020,
+      xlab = "Moose density (moose per km^2)",
```

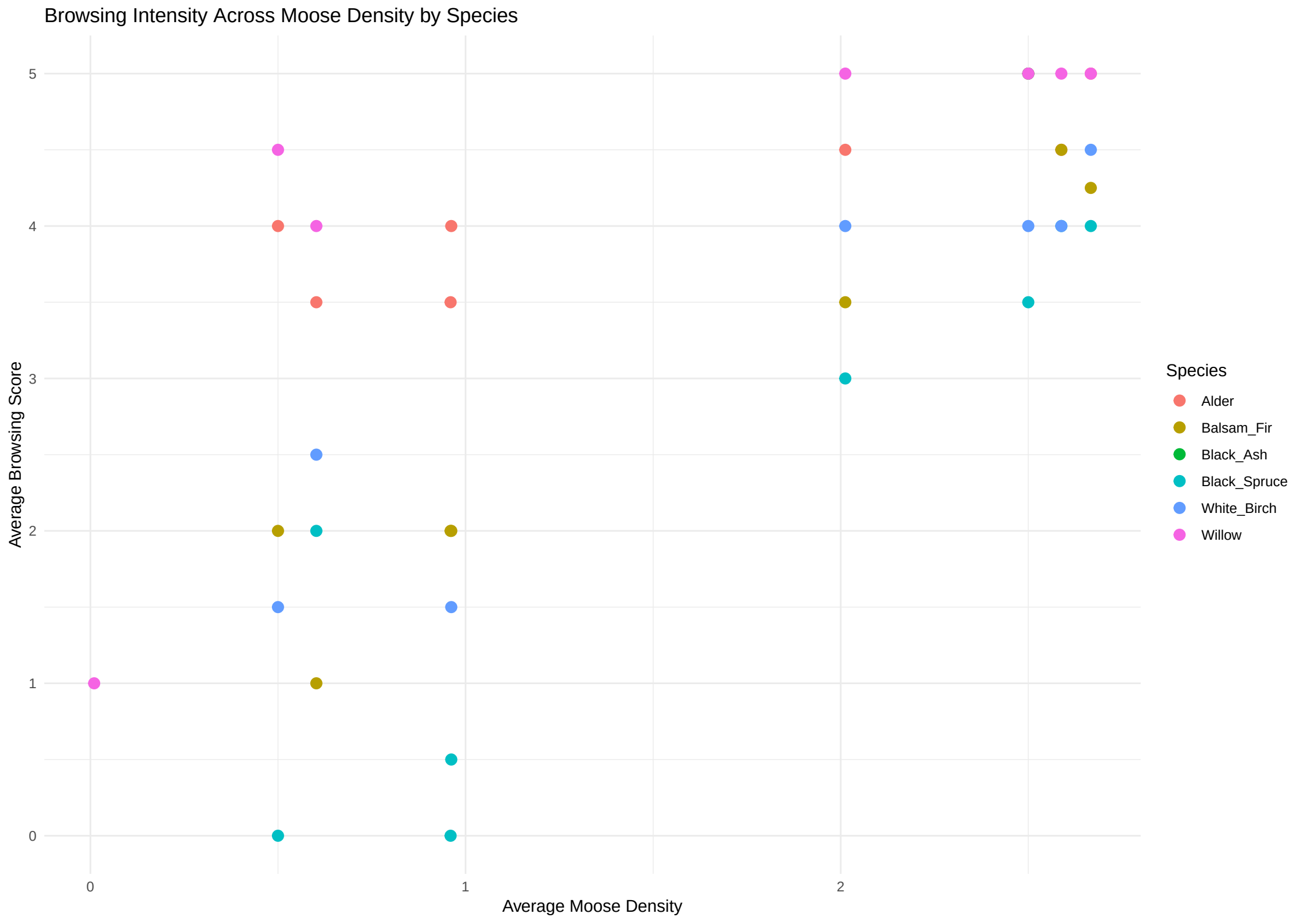
```
+      ylab = "Number of moose-vehicle collisions (2020)",
+      main = "Moose density vs moose-vehicle collisions")
>
> # There is a general positive relationship between moose density and the
number of
> # moose-vehicle collisions, with regions of higher moose density tending
to report
> # more collisions. Avalon_Forests appears to be an outlier, with a much
higher number
> # of collisions than expected relative to its moose density.
> #Q27
> coll_merge_per_capita <- coll_merge %>%
+     mutate(coll_per_capita = collisions2020 / human_pop)
>
> View(coll_merge_per_capita)
> #Q28
> plot(coll_merge_per_capita$human_pop,
+      coll_merge_per_capita$coll_per_capita,
+      xlab = "Human population",
+      ylab = "Moose collisions per capita",
+      main = "Moose-vehicle collisions per capita vs human population")
>
> #29. Moose collisions per capita tend to be higher in regions with small
er human populations and lower in highly populated regions.
> # This pattern makes sense because rural areas have fewer people but mor
e moose habitat and higher exposure to moose on roads, while urban areas d
ilute collision risk across a larger population.
```

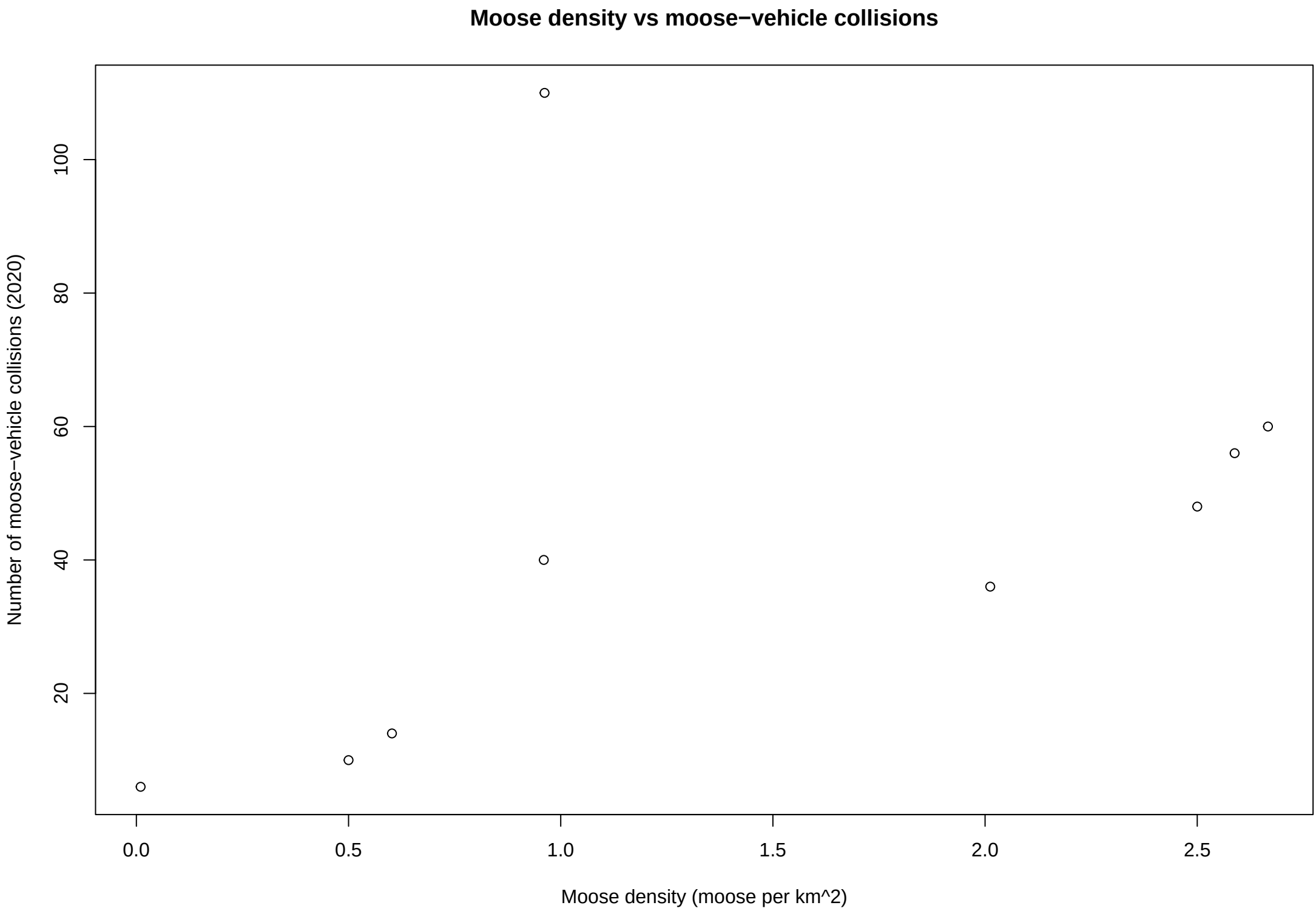

Changes in moose density over time
Moose density (moose per km^2)
0.0
0.5
1.0
1.5
2.0
2.5
1900
1920
1940
1960
1980
2000
2020
Year


Average Balsam Fir browsing intensity by ecoregion
Average browsing score
0
1
2
3
4
Avalon_Forests
Central_Forests
EasternHyperOceanicBarrens
Long_Range_Barrens
Maritime_Barrens
North_Shore_Forests
Northern_Peninsula_Forests
Ecoregion


Average Black Spruce browsing intensity by ecoregion
Average browsing score
0
1
2
3
4
Avalon_Forests
Central_Forests
EasternHyperOceanicBarrens
Long_Range_Barrens
Maritime_Barrens
North_Shore_Forests
Northern_Peninsula_Forests
Western_Forests
Ecoregion


Moose−vehicle collisions per capita vs human population
Moose collisions per capita
0.001
0.002
0.003
0.004
0.005
0
50000
100000
150000
200000
250000
Human population


Browsing Intensity Across Moose Density by Species
Average Browsing Score
Average Moose Density
0
1
2
3
4
5
0
1
2
Species
Alder
Balsam_Fir
Black_Ash
Black_Spruce
White_Birch
Willow


Moose density vs moose−vehicle collisions
Number of moose−vehicle collisions (2020)
20
40
60
80
100
0.0
0.5
1.0
1.5
2.0
2.5
Moose density (moose per km^2)